COSC 39: Theory of Computation

Problem 1. Binary addition. Let's leave the theory land and apply our knowledge in finite automata to some real problems.
For any string $w \in \{0,1\}^*$, let $\langle w \rangle_2$ denote the integer represented by w in binary. For example:

$$\langle \epsilon \rangle_2 = 0, \quad \langle 0011 \rangle_2 = 3, \quad \langle 1010 \rangle_2 = 10, \quad \langle 1111 \rangle_2 = 15.$$

Let L and L' be arbitrary automatic languages over the alphabet $\{0,1\}$. Prove that the following language is also automatic:

$$\{w \in \{0,1\}^* : \langle w \rangle_2 = \langle x \rangle_2 + \langle y \rangle_2 \text{ for some } x \in L, y \in L'\}.$$

Hint: Try to prove first the case when both L and L' are languages containing a single binary integer.

Problem 2. When does an NFA accept everything?. Let N be an arbitrary NFA with n states for some alphabet Σ of size at least 2. How long can the shortest word rejected by N be?
Put differently, let $L(N)$ be the language recognized by the NFA N, and define

$$\Sigma^{\leq f(n)} := \Sigma^0 \cup \cdots \cup \Sigma^{f(n)}.$$

What is the smallest $f(n)$ such that

$$\Sigma^{\leq f(n)} \subseteq L(N) \implies L(N) = \Sigma^*?$$

(a) Prove that $f(n) \leq 2^n$.

(b) Consider the alphabet

$$\Sigma := \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \# \right\}.$$

Define s_n to be the following word:

$$s_n := \# \begin{pmatrix} 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} \cdots \begin{pmatrix} 0 \\ 1 \end{pmatrix} \# \begin{pmatrix} 0 \\ 0 \end{pmatrix} \cdots \begin{pmatrix} 1 \\ 0 \end{pmatrix} \# \cdots \# \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdots \begin{pmatrix} 0 \\ 1 \end{pmatrix} \#.$$

Notice that if we drop the brackets, then we can interpret the 0s and 1s between two consecutive $\#$ symbols (referred to as a *chunk*) as two binary numbers stacked on top of each other.

If we read the top numbers in every chunk from left to right, they form a binary counter from 0 to $2^n - 1$. For the bottom numbers, they form a binary counter from 1 to 2^n.

For example,

$$s_2 := \# \binom{0}{0} \binom{0}{1} \# \binom{0}{1} \binom{1}{0} \# \binom{1}{1} \binom{0}{1} \#,$$

the top numbers are $00, 01, 10$, and the bottom numbers are $01, 10, 11$.

Construct an NFA recognizing the language $\Sigma^* \setminus \{s_n\}$.

Hint: You need to build separate gadgets to check:

- each chunk between two consecutive # has length exactly n,
- the two counts within a chunk differ by one,
- the counts in adjacent chunks are consistent.

For full credit, you can use at most $C \cdot n + o(n)$ states for constant C. This shows $f(n) \geq \Omega(2^n/C)$.

$\star$ (c) Prove or disprove that $f(n) \geq 2^{n-o(n)}$.